higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

9 December 2025 (X-paper)
09:00–12:00

Drawing instruments may be used. Calculators may NOT be used

This question paper consists of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only blue or black ink.
5. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Which of the following is a non-weighed, non-self-complementing code and is used to encode data from a computer?

A 8421 code
B 5421 code
C Grey code
D Excess-3 code

1.1.2 The 5421 BCD code of 7_{10} is …

A 1101
B 1010
C 0111
D 0100

1.1.3 What is the simplest form of the expression $\overline{A} + AB + A$?

A A
B 1
C 0
D $\overline{A}$

1.1.4 What does CMOS stand for?

A Complicated Metal Oxide Semiconductor
B Complementary Metal Oxide Semiconductor
C Current Metal Oxide Sourcing
D Computer Metal Oxide Semiconductor

1.1.5 Which inputs are needed for JK flip-flops to be toggled?

A J = 0 ; K = 0
B J = 1 ; K= 0
C J = 0 ; K = 1
D J = 1 ; K = 1

(5 × 1) (5)

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1.2 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–E) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK

	COLUMN A		COLUMN B
1.2.1	The maximum number of standard inputs that can reliably be driven by one output	A	propagation delay
1.2.2	The time it takes a sequel to move through the system	B	clear
1.2.3	Sending various signal through a single channel	C	fan out
1.2.4	Input that forces the Q output of a memory element to 0	D	fan in
1.2.5	The number of inputs available to be coupled to a specific logic output	E	multiplexer

(5 × 1) (5)
[10]

QUESTION 2

2.1 How many bits does BCD code use to represent each decimal digit? (1)

2.2 Define *self-complementary code*. (2)

2.3 Convert $68,7_{10}$ to Excess-3 code. (3)

2.4 Do the following calculations in the binary number system. Show all conversion and calculations clearly.

2.4.1 $15,6_8 - 12, 125_{10}$
Make use of the complement of two and convert the final answer to a decimal number. (6)

2.4.2 $92,5_{10} \div 3, C_{16}$
Convert the final answer to an octal number. (6)

2.4.3 $6, D_{16} \times 7, 6_8$
Convert the final answer to a hexadecimal number. (6)
[24]

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QUESTION 3

3.1 Study FIGURE 1 below, and answer the following questions:

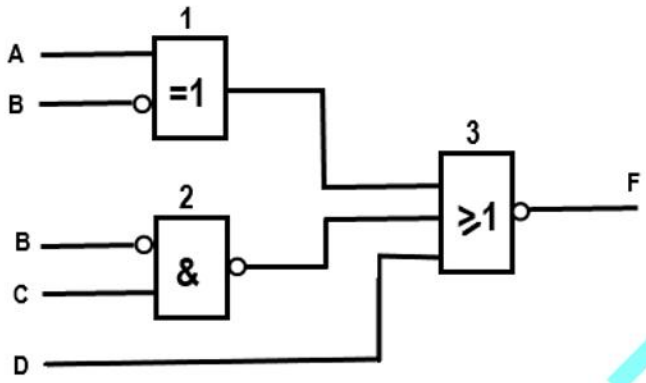


FIGURE 1

3.1.1 What type of gate is represented by gate 1 in FIGURE 1 above? (1)

3.1.2 What is logic expression of gate 2? (1)

3.1.3 Write the logic expression for FIGURE 1 above. (3)

3.2 Simplify the following Boolean expression:

3.2.1 $B\overline{C}\,\overline{D} + \overline{B}\overline{C}D + BCD + \overline{B}C\overline{D} = F$ (6)

3.2.2 $AB + \overline{\overline{A} + B} + \overline{A\overline{B}C + \overline{A}BC} = F$ (5)

3.3 Draw the circuit diagram of the Boolean expression below. Do NOT simplify the expression:

3.3.1 $\overline{A + \overline{B}C \oplus A\overline{C}} = F$ (4)

3.3.2 $\left(X + \overline{Y + XZ}\right)\left(\overline{YZ + X}\right) = F$ (6)

[26]

QUESTION 4

4.1 Draw a neat, labelled diagram of the gated equivalent of a clocked SR flip-flop. Clearly show set, reset and give its truth table. (10)

4.2 Draw a neat, labelled diagram of the ring counter. Also include the output waveform. (10)

[20]

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QUESTION 5

5.1 Draw a neat, labelled diagram of a parallel binary adder. (7)

5.2 Draw a neat, labelled diagram of a 1 × 4 demultiplexer. (7)

5.3 Draw a neat, labelled weighted resistor ladder network digital to analogue (D/A) converter. (6)

[20]

TOTAL: 100

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